

Section 2.1: Matrix Operations

Section 2.2: The Inverse of a Matrix

Section 2.3: Characterizations of Invertible Matrices

The Invertible Matrix Theorem

- Let A be an $n \times n$ matrix. The following are equivalent:
 - a. The matrix A is an invertible matrix.
 - b. The matrix A is row equivalent to the $n \times n$ identity matrix.
 - c. A has n pivot positions.
 - d. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
 - e. The columns of A form a linearly independent set.

The Invertible Matrix Theorem

- f. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
- g. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
- h. The columns of A span $\mathbb{R}^n$.
- i. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- j. There is an $n \times n$ matrix C such that $CA = I$.
- k. There is an $n \times n$ matrix D such that $AD = I$.
- l. A^T is an invertible matrix.

The Invertible Matrix Theorem

- Additionally, whenever A is invertible:

$$A^{-1} \text{ is invertible and } (A^{-1})^{-1} = A$$

- The Invertible Matrix Theorem divides the set of all $n \times n$ matrices into two disjoint classes:

the invertible (nonsingular) matrices, and the noninvertible (singular) matrices.

The Invertible Matrix Theorem

- **Example:** Use the Invertible Matrix Theorem to decide if A is invertible:

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

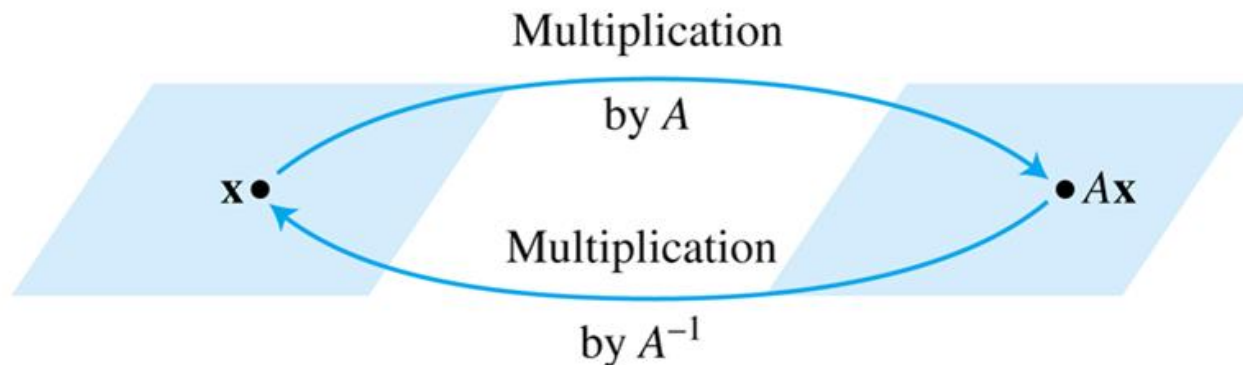
- Solution:

$$A \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

So A has three pivot positions and hence is invertible, by the Invertible Matrix Theorem, statement (c). 

Invertible Linear Transformations

- Matrix multiplication corresponds to composition of linear transformations.
- When a matrix A is invertible, the equation $A^{-1}A\mathbf{x} = \mathbf{x}$ can be viewed as a statement about linear transformations. See the following figure.



A^{-1} transforms $A\mathbf{x}$ back to $\mathbf{x}$.

Invertible Linear Transformations

- A linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is said to be invertible if there exists a function $S: \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that

$$S(T(\mathbf{x})) = \mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^n \quad \text{----(1)}$$

$$T(S(\mathbf{x})) = \mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^n \quad \text{----(2)}$$

- **Theorem:** Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T . Then T is invertible if and only if A is an invertible matrix. In that case, the linear transformation S given by $S(\mathbf{x}) = A^{-1}\mathbf{x}$ is the unique function satisfying equation (1) and (2).

Section 2.4: Partitioned Matrices

Section 2.5: The LU Factorization

Section 2.8: Subspaces of $\mathbb{R}^n$

Section 2.9: Dimension and Rank

Partitioned Matrices

- A key feature of our work with matrices has been the ability to regard matrix A as a list of column vectors rather than just a rectangular array of numbers.
- This point of view has been so useful that we wish to consider other **partitions** of A , indicated by horizontal and vertical dividing rules, as in Example 1 on the next slide.

Partitioned Matrices

- **Example:** The matrix

$$A = \begin{bmatrix} 3 & 0 & -1 & 5 & 9 & -2 \\ -5 & 2 & 4 & 0 & -3 & 1 \\ -8 & -6 & 3 & 1 & 7 & -4 \end{bmatrix}$$

- Can also be written 2×3 **partitioned** (or **block**) **matrix** as the

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

- Whose entries are the **blocks** (or **submatrices**)

$$A_{11} = \begin{bmatrix} 3 & 0 & -1 \\ -5 & 2 & 4 \end{bmatrix}, A_{12} = \begin{bmatrix} 5 & 9 \\ 0 & -3 \end{bmatrix}, A_{13} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$
$$A_{21} = \begin{bmatrix} -8 & -6 & 3 \end{bmatrix}, A_{22} = \begin{bmatrix} 1 & 7 \end{bmatrix}, A_{23} = \begin{bmatrix} -4 \end{bmatrix}$$

Addition and Scalar Multiplication

- If matrices A and B are the same size and are partitioned in **exactly the same way**, then it is natural to make the same partition of the ordinary matrix sum $A + B$.
- In this case, each block of $A + B$ is the (matrix) sum of the corresponding blocks of A and B .
- Multiplication of a partitioned matrix by a scalar is also computed block by block.

Addition and Scalar Multiplication

- Partitioned matrices can be multiplied by the usual row—column rule as if the block entries were scalars, provided that for a product AB , the column partition of A matches the row partition of B .
- **Example:** Let

$$A = \begin{bmatrix} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ 0 & -4 & -2 & 7 & -1 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, B = \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ -1 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}$$

- The 5 columns of A are partitioned into a set of 3 columns and then a set of 2 columns. The 5 rows of B are partitioned in the same way—into a set of 3 rows and then a set of 2 rows.

Multiplication of Partitioned Matrices

$$AB = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} A_{11}B_1 + A_{12}B_2 \\ A_{21}B_1 + A_{22}B_2 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -6 & 2 \\ 2 & 1 \end{bmatrix}$$

- It is important for each smaller product in the expression for AB to be written with the submatrix from A on the left, since **matrix multiplication is not commutative**.

Multiplication of Partitioned Matrices

- For instance,

$$A_{11}B_1 = \begin{bmatrix} 2 & -3 & 1 \\ 1 & 5 & -2 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix}$$

$$A_{12}B_2 = \begin{bmatrix} 0 & -4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix}$$

- Hence the top block in AB is

- $A_{11}B_1 + A_{12}B_2 = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix} + \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix} = \begin{bmatrix} -5 & -4 \\ -6 & 2 \end{bmatrix}$

Multiplication of Partitioned Matrices

- **Theorem:** Column—Row Expansion of AB

If A is $m \times n$ And B is $n \times p$, then

$$\begin{aligned} AB &= \begin{bmatrix} \text{col}_1(A) & \text{col}_2(A) & \dots & \text{col}_n(A) \end{bmatrix} \begin{bmatrix} \text{row}_1(B) \\ \text{row}_2(B) \\ \vdots \\ \text{row}_n(B) \end{bmatrix} \\ &= \text{col}_1(A) \text{row}_1(B) + \dots + \text{col}_n(A) \text{row}_n(B) \end{aligned} \quad (1)$$

- **Proof:** For each row index i and column index j , the (i, j) -entry in $\text{col}_k(A)$ and b_{kj} from $\text{row}_k(B)$ is the product of a_{ik} from $\text{col}_k(A)$ and b_{kj} from $\text{row}_k(B)$.

Multiplication of Partitioned Matrices

- Hence the (i, j) -entry in the sum shown in equation (1) is

$$\begin{array}{ccccccc} a_{i1}b_{1j} & + & a_{i2}b_{2j} & + & \cdots & + & a_{in}b_{nj} \\ (k=1) & & (k=2) & & & & (k=n) \end{array}$$

- This sum is also the (i, j) -entry in AB , by the row—column rule.